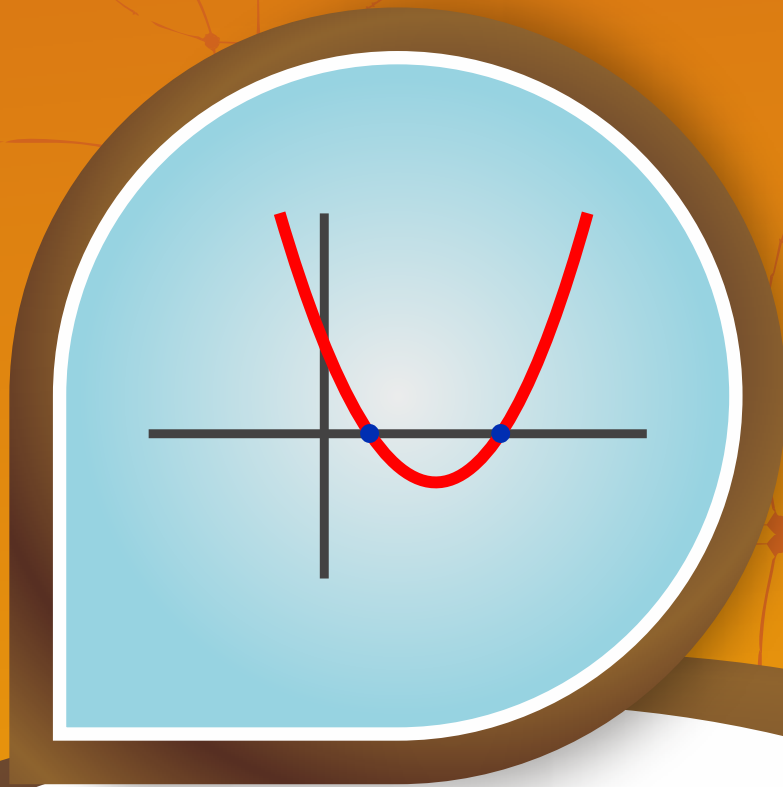


MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Quadratic Equation

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. If the roots of the equation $x^2 - 5x + 16 = 0$ are α , β and the roots of the equation $x^2 + px + q = 0$ are $(\alpha^2 + \beta^2)$ and $\frac{\alpha\beta}{2}$, then -
 (A) $p = 1$ and $q = 56$ (B) $p = 1$ and $q = -56$
 (C) $p = -1$ and $q = 56$ (D) $p = -1$ and $q = -56$
QE0001
2. If the roots of the equation $x^2 + px + q = 0$ are 8 and 2 and the roots of $x^2 + rx + s = 0$ are 3 and 3, then roots of $x^2 + px + s = 0$ are
 (A) -1, -9 (B) 1, 9 (C) 8, 3 (D) None
QE0002
3. If α and β be the roots of the equation $(x - a)(x - b) = c$ and $c \neq 0$, then roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are -
 (A) a and c (B) b and c (C) a and b (D) $a + b$ and $b + c$
QE0003
4. If $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$ then the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is -
 (A) $19/3$ (B) $25/3$ (C) $-19/3$ (D) none of these
QE0004
5. The value of a for which one roots of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is
 (A) $-2/3$ (B) $1/3$ (C) $-1/3$ (D) $2/3$
QE0005
6. The number of real solutions of the equation $x^2 - 3|x| + 2 = 0$, is-
 (A) 4 (B) 1 (C) 3 (D) 2
QE0006
7. If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its roots are-
 (A) 0, -1 (B) -1, 1 (C) 0, 1 (D) -1, 2
QE0007
8. If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals -
 (A) 1 (B) 2 (C) 3 (D) -2
QE0008
9. If $x^2 - A(x + 1) + C = 0$ has roots x_1 & x_2 , then the value of $x_1^2 + x_2^2 + (2 + A)x_1x_2$, is-
 (A) AC (B) $A^2 + AC$ (C) $A^2 - AC$ (D) $-AC$
QE0009
10. The quadratic $x^2 + ax + b + 1 = 0$ has roots which are positive integers, then $(a^2 + b^2)$ can be equal to-
 (A) 50 (B) 17 (C) 29 (D) 53
QE0010

11. **Statement-1** : If sum of the roots of quadratic equation $2x^2 + bx + c = 0$ is equal to sum of the squares of the roots, then $b^2 + 2b = 4c$.

Statement-2 : If one root of quadratic equation $x^2 + px + q = 0$, is $4 + \sqrt{3}$, then other root is $4 - \sqrt{3}$.

- (A) Statement-1 is true and Statement-2 is false.
 (B) Statement-1 and Statement-2 both are true.
 (C) Statement-1 is false and Statement-2 is true.
 (D) Statement-1 and Statement-2 both are false.

QE0011

12. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of 'q' is-

- (A) 3 (B) 12 (C) 49/4 (D) 4

QE0012

13. The sum of the values of m for which the quadratic polynomial $P(x) = x^2 + (m + 5)x + (5m + 1)$ is a perfect square of linear in 'x' ($m \in \mathbb{R}$) is

- (A) 3 (B) 7 (C) 8 (D) 10

QE0013

14. If equations $x^2 - 5x + 5 = 0$ and $x^3 + ax^2 + bx + 5 = 0$ have common root, then value of $a + b$ ($a, b \in \mathbb{Q}$) is -

- (A) 4 (B) -4 (C) 0 (D) can't find

QE0014

15. If the equation $ax^2 + bx + c = 0$ has distinct real roots, both negative, then-

- (A) a,b,c must be of same sign
 (B) a,b must be of opposite sign
 (C) a,c must be of opposite sign
 (D) a,b must be of same sign and opposite to sign of c

QE0015

16. If $P(x) = x^2 - (2 - p)x + p - 2$ assumes both positive and negative value, then the complete set of values of 'p' is -

- (A) $(-\infty, 2)$ (B) $(6, \infty)$ (C) $(2, 6)$ (D) $(-\infty, 2) \cup (6, \infty)$

QE0016

17. If value of a for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assume the least value is -

- (A) 2 (B) 3 (C) 0 (D) 1

QE0017

18. If $x^2 + 2ax + 10 - 3a > 0$ for all $x \in \mathbb{R}$, then

- (A) $-5 < a < 2$ (B) $a < -5$ (C) $a > 5$ (D) $2 < a < 5$

QE0018

19. If the expression $y = 8x - x^2 - 15$ is negative, then x lies in the interval -
 (A) (3,5) (B) (5,50) (C) (3,∞) (D) $(-\infty, 3) \cup (5, \infty)$ QE0019
20. Let $f(x) = ax^2 + bx + 8$ ($a, b \in \mathbb{R}$) be a quadratic polynomial whose graph is symmetric about the line $x = 2$. If minimum value of $f(x)$ is 6, then the value of $2a - b$ is -
 (A) 0 (B) 1 (C) 2 (D) 3 QE0020
21. Let $g(x) = x^2 - (b + 1)x + (b - 1)$, where b is a real parameter. The largest natural number b satisfying $g(x) > -2 \forall x \in \mathbb{R}$, is -
 (A) 1 (B) 2 (C) 3 (D) 4 QE0021
22. $y = x^2 - 6x + 5$, $x \in [2, 4]$, then -
 (A) least value of y is -3 (B) least value of y is 3
 (C) greatest value of y is 4 (D) greatest value of y is -3 QE0022
23. Range of the expression $\frac{16x^2 - 12x + 9}{16x^2 + 12x + 9} : (x \in \mathbb{R})$ is-
 (A) $\left[\frac{1}{3}, 3\right]$ (B) $\left(-\infty, \frac{1}{3}\right]$ (C) $[3, \infty)$ (D) $\mathbb{R}$ QE0023
24. If the roots of equation $(4p - p^2 - 5)x^2 - (2p - 1)x + 3p = 0$ lie on either side of unity then the number of integral values of p is-
 (A) 4 (B) 2 (C) 3 (D) 1 QE0024
25. If exactly one root of the equation $2^k x^2 - 4^k x + 2^k - 1 = 0$ lies in $[0, 1)$, then complete range of k is-
 (A) $(-\infty, 0]$ (B) $(-\infty, 0)$ (C) $(0, \infty)$ (D) $[0, \infty)$ QE0025
26. If $\forall p \in \mathbb{R}$ one root of the equation $x^2 + 2px + q^2 - p^2 - 6 = 0$ is less than 1 and other root is greater than 1, then range of q is -
 (A) $(-\infty, -2)$ (B) $(-2, 2)$ (C) $(-\sqrt{5}, \sqrt{5})$ (D) $(2, \infty)$ QE0026
27. Let $f(x) = 2x^2 + px + 1$ is given. If $f(x)$ is negative integer for only one real value of x , then product of all possible values of p is -
 (A) -3 (B) -16 (C) 5 (D) -7 QE0027

28. Let r_1, r_2, r_3 be roots of equation $x^3 - 2x^2 + 4x + 5074 = 0$, then the value of $(r_1 + 2)(r_2 + 2)(r_3 + 2)$ is
(A) 5050 (B) -5050 (C) -5066 (D) -5068

QE0028

29. Let a, b, c are roots of equation $x^3 + 8x + 1 = 0$, then the value of

$$\frac{bc}{(8b+1)(8c+1)} + \frac{ac}{(8a+1)(8c+1)} + \frac{ab}{(8a+1)(8b+1)} \text{ is equal to}$$

- (A) 0 (B) -8 (C) -16 (D) 16

QE0029

30. Let $f(x) = x^3 + x + 1$ and $P(x)$ be a cubic polynomial such that $P(0) = -1$ and the roots of $P(x) = 0$ are the squares of the roots of $f(x) = 0$, then value of $P(9)$ is -
(A) 98 (B) 899 (C) 80 (D) 898

QE0030

EXERCISE (O-2)

Multiple Correct Answer Type

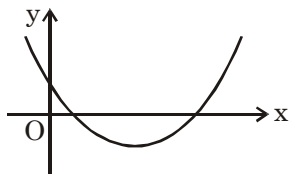
- If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009}$ is greater than or equal to
(A) -2 (B) -1 (C) 1 (D) 2
QE0031
- The real value(s) of x satisfying the equation $2^{(x-1)(x^2+5x-50)} = 1$ is/are :
(A) 1 (B) -5 (C) -10 (D) 5
QE0032
- The value(s) of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is/are :
(A) $\frac{1}{2}$ (B) 0 (C) 3 (D) 1
QE0033
- The integral value(s) of α such that equation $x^2 - \alpha x + \alpha + 1 = 0$ has integral roots, is/are -
(A) -1 (B) 4 (C) 3 (D) 5
QE0034
- For the equation, $3x^2 + px + 3 = 0$, if one of the roots is square of the other, then the value(s) of p is/are -
(A) $1/3$ (B) 1 (C) 3 (D) -6
QE0035
- Let $p, q \in \mathbb{Q}$. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then :
(A) $q^2 + 4p + 14 = 0$ (B) $p^2 - 4q - 12 = 0$ (C) $p = -4, q = 1$ (D) $p = 1, q = -4$
QE0038
- If the equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root different from -1, then the value(s) of 'b' is/are :-
(A) 0 (B) $\sqrt{3}i$ (C) $-\sqrt{3}i$ (D) 2
QE0041
- Let $p(x)$ be a quadratic polynomial such that $p(0) = 1$. If $p(x)$ leaves remainder 4 when divided by $x - 1$ and it leaves remainder 6 when divided by $x + 1$; then :
(A) $p(2) = 19$ (B) $p(-2) = 19$ (C) $p(-2) = 11$ (D) $p(2) = 15$
QE0042
- If α & β ($\alpha < \beta$), are the roots of the equation, $x^2 + bx + c = 0$, where $c < 0 < b$, then -
(A) $\alpha < 0$ (B) $\beta < |\alpha|$ (C) $\beta > 0$ (D) $\beta > |\alpha|$
QE0046

10. If $b > a$ and α, β ($\alpha < \beta$) are the roots of the equation, $(x - a)(x - b) - 1 = 0$, then -

- (A) $\alpha \in [b, \infty)$ (B) $\alpha \in (-\infty, a)$
 (C) $\beta \in (b, \infty)$ (D) $\alpha \in [a, b]$ & $\beta \in [a, b]$

QE0047

11. The graph of $y = ax^2 + bx + c$ is shown. Which of the following hold(s) good ?



- (A) $ab^2c^3 > 0$ (B) $ab^3c^2 < 0$ (C) $ab^3c^5 > 0$ (D) $b^2 > 4ac$

QE0048

12. The value(s) of x which satisfy the equation $|\sqrt{x} - 2| + \sqrt{x}(\sqrt{x} - 4) + 2 = 0$, ($x > 0$), is/are :

- (A) 1 (B) 9 (C) 10 (D) 12

QE0049

13. The subset of the set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is/are -

- (A) $(-\infty, -2) \cup (2, \infty)$ (B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
 (C) $(-\infty, -1) \cup (1, \infty)$ (D) $(\sqrt{2}, \infty)$

QE0050

14. For the equation $|x|^2 + |x| - 6 = 0$, the correct statement (s) is (are) :

- (A) sum of roots is 0 (B) product of roots is -4
 (C) there are 4 real roots (D) there are only 2 real roots

QE0171

15. If α, β are the roots of $ax^2 + bx + c = 0$, and $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0$, (where $h \neq 0$), then

- (A) $\frac{a}{p} = \frac{b}{q} = \frac{c}{r}$ (B) $h = \frac{1}{2} \left(\frac{b}{a} - \frac{q}{p} \right)$ (C) $h = \frac{1}{2} \left(\frac{b}{a} + \frac{q}{p} \right)$ (D) $\frac{b^2 - 4ac}{a^2} = \frac{q^2 - 4pr}{p^2}$

QE0172

16. If a, b are non-zero real numbers and α, β the roots of $x^2 + ax + b = 0$, then

- (A) α^2, β^2 are the roots of $x^2 - (2b - a^2)x + a^2 = 0$

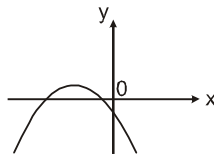
- (B) $\frac{1}{\alpha}, \frac{1}{\beta}$ are the roots of $bx^2 + ax + 1 = 0$

- (C) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are the roots of $bx^2 + (2b - a^2)x + b = 0$

- (D) $(\alpha - 1), (\beta - 1)$ are the roots of the equation $x^2 + x(a + 2) + 1 + a + b = 0$

QE0173

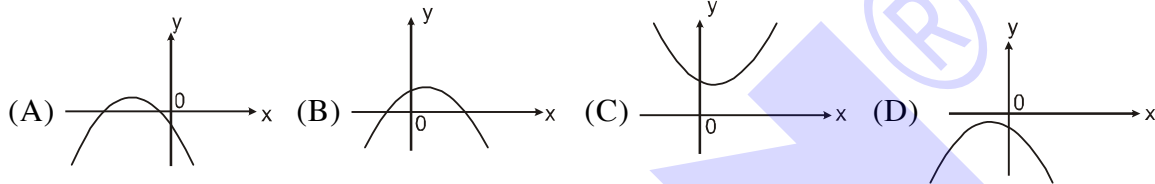
17. The graph of the quadratic polynomial $y = ax^2 + bx + c$ is as shown in the figure. Then :



- (A) $b^2 - 4ac > 0$ (B) $b < 0$ (C) $a > 0$ (D) $c < 0$

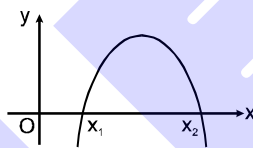
QE0174

18. For which of the following graphs of the quadratic expression $y = ax^2 + bx + c$, the product $a b c$ is negative ?



QE0175

19. The adjoining figure shows the graph of $y = ax^2 + bx + c$. Then



- (A) $a < 0$ (B) $b^2 < 4ac$
(C) $c > 0$ (D) a and b are of opposite sign

QE0176

20. $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d = 0$, then the real root of above equation is
($a, b, c, d \in \mathbb{R}$)

- (A) $-d/a$ (B) d/a (C) $(b - a)/a$ (D) $(a - b)/a$

QE0177

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Consider the expression $y = ax^2 + bx + c$, $a \neq 0$ and $a, b, c \in \mathbb{R}$ then the graph between x , y is always a parabola. If $a > 0$ then the shape of the parabola is concave upward and if $a < 0$ then the shape of the parabola is concave down ward. If $y > 0$ or $y < 0$ then discriminant $D < 0$.

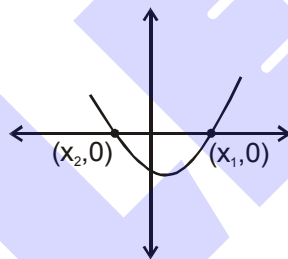
1. Let $x^2 + 2ax + 10 - 3a > 0$ for every real value of x , then –
 (A) $a > 5$ (B) $a < -5$ (C) $-5 < a < 2$ (D) $2 < a < 5$

QE0178

2. The value of $x^2 + 2bx + c$ is positive if –
 (A) $b^2 - 4c > 0$ (B) $b^2 - 4c < 0$ (C) $c^2 < b$ (D) $b^2 < c$

QE0179

3. The diagram show the graph of $y = ax^2 + bx + c$ then –



- (A) $a < 0$ (B) $c < 0$ (C) $b^2 - 4ac < 0$ (D) $b^2 - 4ac = 0$

QE0180

Paragraph for Question 4 to 6

Consider the equation $x^4 - \lambda x^2 + 9 = 0$.

4. If the equation has four real and distinct roots, then the possible value(s) of λ is/are
 (A) 7 (B) -7 (C) 8 (D) 6

QE0181

5. If the equation has no real root, then the possible value(s) of λ is/are
 (A) 6 (B) -7 (C) 0 (D) -6

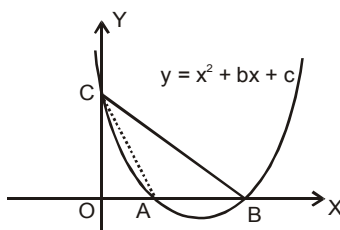
QE0182

6. If the equation has only two real roots, then which of the following value(s) of λ is/are Not possible
 (A) 7 (B) -7 (C) 0 (D) 6

QE0183

Paragraph for Question 7 and 8

In the given figure $\triangle OBC$ is an isosceles right triangle in which AC is a median, then answer the following questions :



7. Sum of roots of $y = 0$ is

QE0184

8. Minimum value of the quadratic expression corresponding to the quadratic equation whose roots are $(\alpha + \beta)$ & $(\alpha - \beta)$, where α, β ($\alpha > \beta$) are roots obtained in previous question occurs at $x =$

QE0185

Matrix Match Type

9.	List-I	List-II
(P)	If $\alpha, \alpha + 4$ are two roots of $x^2 - 8x + k = 0$, then possible value of k is	(1) 2
(Q)	Number of real roots of equation $x^2 - 5 x + 6 = 0$ are 'n', then value of $\frac{n}{2}$ is	(2) 3
(R)	If $3 - i$ is a root of $x^2 + ax + b = 0$ ($a, b \in \mathbb{R}$), then b is	(3) 12
(S)	If both roots of $x^2 - 2kx + k^2 + k - 5 = 0$ are less than 5, then 'k' may be equal to	(4) 10
(A)	(P) $\rightarrow$ (3), (Q) $\rightarrow$ (1), (R) $\rightarrow$ (4), (S) $\rightarrow$ (2)	
(B)	(P) $\rightarrow$ (1), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (4)	
(C)	(P) $\rightarrow$ (3), (Q) $\rightarrow$ (2), (R) $\rightarrow$ (3), (S) $\rightarrow$ (2)	
(D)	(P) $\rightarrow$ (4), (Q) $\rightarrow$ (3), (R) $\rightarrow$ (2), (S) $\rightarrow$ (1)	

QE0186

10. Let $f(x) = \frac{x^2 - 6x + 5}{x^2 - 5x + 6}$

Match the expressions/statements in Column I with expressions/statements in Column II.

Column-I	Column-II
(A) If $-1 < x < 1$, then $f(x)$ satisfies	(p) $0 < f(x) < 1$
(B) If $1 < x < 2$, then $f(x)$ satisfies	(q) $f(x) < 0$
(C) If $3 < x < 5$, then $f(x)$ satisfies	(r) $f(x) > 0$
(D) If $x > 5$, then $f(x)$ satisfies	(s) $f(x) < 1$

QE0054

EXERCISE (O-4)

Numerical Grid Type

1. α, β are the roots of the equation $K(x^2 - x) + x + 5 = 0$. If K_1 & K_2 are the two values of K for which the roots α, β are connected by the relation $(\alpha/\beta) + (\beta/\alpha) = 4/5$. Find the value of $(K_1/K_2) + (K_2/K_1)$.

QE0055

2. Let the quadratic equation $x^2 + 3x - k = 0$ has roots a, b and $x^2 + 3x - 10 = 0$ has roots c, d such that modulus of difference of the roots of the first equation is equal to twice the modulus of the difference of the roots of the second equation. If the value of 'k' can be expressed as rational number in the lowest form as m/n then find the value of $(m + n)$.

QE0056

3. The sum of all possible value(s) of m for which the quadratic equations $x^2 - 11x + m = 0$ and $x^2 - 14x + 2m = 0$ may have common root, is

QE0059

4. Let a, b be arbitrary real numbers. Find the smallest natural number 'b' for which the equation $x^2 + 2(a + b)x + (a - b + 8) = 0$ has unequal real roots for all $a \in \mathbb{R}$.

QE0060

5. Consider the quadratic polynomial $f(x) = x^2 - 4ax + 5a^2 - 6a$. Find the smallest positive integral value of 'a' for which $f(x)$ positive for every real x .

QE0062

6. We call 'p' a good number if the inequality $\frac{2x^2 + 2x + 3}{x^2 + x + 1} \leq p$ is satisfied for any real x . Find the smallest integral good number.

QE0063

7. Number of integral values of 'a' for which $2x^2 - 2ax + a^2 - a - 6 = 0$ has roots of opposite sign is

QE0064

8. If '4' lies between the roots of the equation $x^2 - (3k - 1)x + 5k = 0$, then minimum possible integral value of k is

QE0187

9. Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find the value of $(\alpha - 2)(\beta - 2)(\gamma - 2)$.

QE0188

10. Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$

QE0073

EXERCISE (JM)

1. Let for $a \neq a_1 \neq 0$, $f(x) = ax^2 + bx + c$, $g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$.
If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is : [AIEEE-2011]
(1) 18 (2) 3 (3) 9 (4) 6
QE0081
2. Sachin and Rahul attempted to solve a quadratic equation. Sachin made a mistake in writing down the constant term and ended up in roots (4, 3). Rahul made a mistake in writing down coefficient of x to get roots (3, 2). The correct roots of equation are : [AIEEE-2011]
(1) -4, -3 (2) 6, 1 (3) 4, 3 (4) -6, -1
QE0082
3. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ is : [JEE-MAIN-2013]
(1) 1 : 2 : 3 (2) 3 : 2 : 1 (3) 1 : 3 : 2 (4) 3 : 1 : 2
QE0083
4. Let α and β be the roots of equation $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$, for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is equal to : [JEE-MAIN-2015]
(1) 3 (2) -3 (3) 6 (4) -6
QE0084
5. The sum of all real values of x satisfying the equation $(x^2 - 5x + 5)^{x^2 + 4x - 60} = 1$ is :- [JEE-MAIN-2016]
(1) 5 (2) 3 (3) -4 (4) 6
QE0085
6. Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to : [JEE(Main)-2019]
(1) 512 (2) -512 (3) -256 (4) 256
QE0086
7. The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is : [JEE(Main)-2019]
(1) 2 (2) 5 (3) 3 (4) 4
QE0087

8. Consider the quadratic equation $(c - 5)x^2 - 2cx + (c - 4) = 0$, $c \neq 5$. Let S be the set of all integral values of c for which one root of the equation lies in the interval (0,2) and its other root lies in the interval (2,3). Then the number of elements in S is : [JEE(Main)-2019]

(1) 11 (2) 18 (3) 10 (4) 12

QE0088

9. If λ be the ratio of the roots of the quadratic equation in x, $3m^2x^2 + m(m-4)x + 2 = 0$, then the least value of m for which $\lambda + \frac{1}{\lambda} = 1$, is : [JEE(Main)-2019]

(1) $2 - \sqrt{3}$ (2) $4 - 3\sqrt{2}$ (3) $-2 + \sqrt{2}$ (4) $4 - 2\sqrt{3}$

QE0089

10. Let α and β be the roots of the equation $x^2 - x - 1 = 0$. If $p_k = (\alpha)^k + (\beta)^k$, $k \geq 1$, then which one of the following statements is not true ? [JEE (Main)-2020]

(1) $(p_1 + p_2 + p_3 + p_4 + p_5) = 26$ (2) $p_5 = 11$
(3) $p_3 = p_5 - p_4$ (4) $p_5 = p_2 \cdot p_3$

QE0090

11. The least positive value of 'a' for which the equation $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is

[JEE(Main)-2020]

QE0091

12. If $A = \{x \in \mathbf{R} : |x| < 2\}$ and $B = \{x \in \mathbf{R} : |x - 2| \geq 3\}$; then :

(1) $A \cup B = \mathbf{R} - (2, 5)$ (2) $A \cap B = (-2, -1)$
(3) $B - A = \mathbf{R} - (-2, 5)$ (4) $A - B = [-1, 2)$

[JEE(Main)-2020]

QE0092

13. Let p and q be two positive numbers such that $p + q = 2$ and $p^4 + q^4 = 272$. Then p and q are roots of the equation : [JEE(Main)-2021]

(1) $x^2 - 2x + 2 = 0$ (2) $x^2 - 2x + 8 = 0$
(3) $x^2 - 2x + 136 = 0$ (4) $x^2 - 2x + 16 = 0$

QE0189

14. The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots}}}}$ is equal to [JEE(Main)-2021]

(1) $1.5 + \sqrt{3}$ (2) $2 + \sqrt{3}$ (3) $3 + 2\sqrt{3}$ (4) $4 + \sqrt{3}$

QE0190

15. The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is :

[JEE(Main)-2022]

- (A) 4 (B) 5 (C) 6 (D) 8

QE0191

16. Let $S = \left\{ x \in [-6, 3] - \{-2, 2\} : \frac{|x+3|-1}{|x|-2} \geq 0 \right\}$ and $T = \{x \in \mathbb{Z} : x^2 - 7|x| + 9 \leq 0\}$. Then the number of elements in $S \cap T$ is

[JEE(Main)-2022]

- (A) 7 (B) 5 (C) 4 (D) 3

QE0192

17. The number of integral values of k , for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other root lies in the interval $(2, 3)$, is :

[JEE(Main)-2023]

- (1) 2 (2) 0 (3) 1 (4) 3

QE0193

18. Let $S = \left\{ \alpha : \log_2(9^{2\alpha-4} + 13) - \log_2\left(\frac{5}{2} \cdot 3^{2\alpha-4} + 1\right) = 2 \right\}$. Then the maximum value of β for which the

equation $x^2 - 2\left(\sum_{\alpha \in S} \alpha\right)^2 x + \sum_{\alpha \in S} (\alpha + 1)^2 \beta = 0$ has real roots, is _____.

[JEE(Main)-2023]

QE0194

EXERCISE (JA)

1. The smallest value of k , for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is [JEE 2009, 4 (-1)]

QE0093

2. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are nonzero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is [JEE 2010, 3]

- (A) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$ (B) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
 (C) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$ (D) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

QE0094

3. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is [JEE 2011]

- (A) 1 (B) 2 (C) 3 (D) 4

QE0095

4. A value of b for which the equations

$$x^2 + bx - 1 = 0$$

$$x^2 + x + b = 0,$$

have one root in common is -

[JEE 2011]

- (A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

QE0096

5. Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is(are) a subset(s) of S ?

[JEE 2015, 4M, -0M]

- (A) $\left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}}, 0\right)$ (C) $\left(0, \frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

QE0097

PARAGRAPH

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0, 1, 2, \dots$, let $a_n = p\alpha^n + q\beta^n$.

FACT : If a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$.

6. If $a_4 = 28$, then $p + 2q =$

(A) 14

(B) 7

(C) 12

(D) 21

[JEE(Advanced)-2017, 3(-1)]

QE0098

7. $a_{12} =$

(A) $2a_{11} + a_{10}$

(B) $a_{11} - a_{10}$

(C) $a_{11} + a_{10}$

(D) $a_{11} + 2a_{10}$

[JEE(Advanced)-2017, 3(-1)]

QE0098

8. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is

[JEE(Advanced)-2020]

(A) 0

(B) 8000

(C) 8080

(D) 16000

QE0169

9. For $x \in \mathbb{R}$, then number of real roots of the equation $3x^2 - 4|x^2 - 1| + x - 1 = 0$ is ____.

[JEE(Advanced)-2021]

QE0170

ANSWER KEY

Do yourself-1

- 1 (a) $-1, -2$; (b) 4 ; (c) $1 \pm \sqrt{2}$;
 2 $a, \frac{1}{a}$; 3. $\frac{7}{3}$ 4. $3, -\frac{1}{5}$
 5. $28x^2 - 20x + 1 = 0$ 6. $a_1 = -2, a_2 = 1$
 7. $a_1 = 2, a_2 = 9/2$ 8. $p = \pm 7$
 9. $k = \pm 3\sqrt{5}$ 10. $p = 0$
 11. $a = 2$ 12. $x_1^3 + x_2^3 = 3pq - p^3$
 13. $x_1^3 + x_2^3 = \frac{a(a^2 - 18a + 9)}{27}$ 14. $\frac{215}{27}$

Do yourself-2

1. (a) imaginary; (b) real & distinct; (c) real & coincident
 3. $a = 20 \pm 6\sqrt{5}$ 4. $k_1 = \frac{-22}{3}, k_2 = 2$ 5. $a = 4$
 6. $a = \pm 2$ 7. $k = 3$ 9. For all $m \in \left(\frac{1}{4}, +\infty\right)$
 10. For all $c \in [2, 4)$ 11. $k = 13$ 12. For all $m \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$
 13. For all $m \in \left(-\infty, -\frac{1}{7}\right) \cup (1, \infty)$ 14. $b = -4, c = 1$;
 15. 7 16. -15
 17. (a) $c = 0$; (b) $c = 1$; (c) $b \rightarrow \text{negative}, c \rightarrow \text{negative}$ 18. (ii), (iii) and (iv)
 19. (A) $\frac{b^2 - 2ac}{c^2}$ (B) $\frac{bc^4(3ac - b^2)}{a^7}$ (C) $\frac{b^2(b^2 - 4ac)}{a^2c^2}$
 20. $\frac{1}{x_1^3} + \frac{1}{x_2^3} = -\frac{27a^3 + 36a}{8}$ 21. $a_1 = 3/2, a_2 = 3$ 22. $a = -4$
 23. $a_1 = -\frac{3}{2}, a_2 = 6$. 24. $\{2, 18\}$ for $a = 6, \left\{\frac{2}{19}, \frac{18}{19}\right\}$ for $a = -\frac{6}{19}$
 25. $a_1 = -\frac{125}{8}, a_2 = \frac{27}{8}$ 26. $a_1 = 1/2, a_2 = 1$ 27. $p_1 = 0, q_1 = 0, p_2 = 1, q_2 = -2$

Do yourself-3

- (a) $c^2y^2 + y(2ac - b^2) + a^2 = 0$; (b) $acx^2 - bx + 1 = 0$;
(c) $acx^2 + (a + c)bx + (a + c)^2 = 0$
- $x^2 - 2(p^2 - 2q)x + p^2(p^2 - 4q) = 0$
- $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

Do yourself-4

- 1
- $b = \frac{9}{2}, c = 5$
- $c = 0, c = 6$
- $a = -2$
- $m = -2$

Do yourself-5

- $x \in \mathbb{R}$
- $x \in \mathbb{R}$
- $x \in \left(-\infty, \frac{1}{2}\right) \cup (3, \infty)$
- $x \in (-6, -3) \cup \left(\frac{1}{2}, 2\right) - \{1\} \cup (9, \infty)$
- $[3, 7]$
- $x \in \mathbb{R}$
- $x \in (-\infty, -3) \cup (-2, 3)$
- $\left(\frac{1}{2}, 3\right)$
- $(2, 3)$
- $(-\infty, -2) \cup \left(-2, -\frac{1}{2}\right) \cup (1, \infty)$

Do yourself-6

- $x \in \mathbb{R} - (0, 1]$
- $(1/2, 2]$
- $(3, +\infty)$
- $[1, 2) \cup (2, +\infty)$
- $[20/9, 4) \cup (5, +\infty)$
- $\left(\frac{1}{2}, \frac{5}{2}\right]$
- $[0, 2]$
- $x \in [-18, -2)$

Do yourself-7

- $(-\infty, -10] \cup [10, \infty)$
- $x \in [0, 10]$
- $x \in [-2, 6]$
- $\left[-\frac{7}{2}, 1\right] \cup \left[2, \frac{13}{2}\right]$
- $(-\sqrt{2}, \sqrt{2})$
- ϕ
- $\left(-\frac{3}{2}, \frac{3}{2}\right)$
- ϕ
- $\left(1, \frac{7}{3}\right)$
- $[-1, 3]$
- $(-\infty, -2) \cup (2, \infty)$
- $\left(-\frac{6}{7}, \frac{10}{7}\right)$
- $x \in \mathbb{R}$
- $\left(-\infty, -\frac{7}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$

Do yourself-8

1. (a) 1 (b) -1
2. (1) (i) $a < 0$ (ii) $b < 0$ (iii) $c < 0$ (iv) $D > 0$ (v) $\alpha + \beta < 0$ (vi) $\alpha\beta > 0$
 (2) (i) $a < 0$ (ii) $b > 0$ (iii) $c = 0$ (iv) $D > 0$ (v) $\alpha + \beta > 0$ (vi) $\alpha\beta = 0$
 (3) (i) $a < 0$ (ii) $b = 0$ (iii) $c = 0$ (iv) $D = 0$ (v) $\alpha + \beta = 0$ (vi) $\alpha\beta = 0$
3. Third quadrant 4. (a) $a > 9/16$ (b) $a < -2$ 5. $c < 0$
6. For all $a \in (-6, 3)$ 7. $(-3, 5)$ 8. $k = 3$
9. $\left[0, \frac{1}{2}\right)$ 10. $k = 5$ 11. For all $a \in (6, +\infty)$

Do yourself-9

1. $(-\infty, 2] \cup [4, \infty)$ 2. $[0, 1)$ 3. $\left[\frac{1}{3}, 3\right]$ 4. $\left[-\frac{1}{11}, 1\right]$
5. $(-\infty, 5] \cup [9, \infty)$ 6. $\frac{1}{3}$

Do yourself-10

1. (a) α and β are greater than -3
 (b) '0' lies in between the roots α and β .
 (c) α and β are less than 1.
2. C 3. $a < 2$ 4. $a < 2$ 5. $12 < \lambda < 16$
6. For all $a \in \left(2, \frac{9}{4}\right]$

Do yourself-11

1. 0, 2 2. -2 3. ± 7
4. $(n\ell' - n'\ell)^2 = (mn' - m'n)(\ell m' - \ell'm)$
5. $(aa' - bb')^2 = -4(ah' + b'h)(a'h + bh')$

Do yourself-12

2. -27, 5
3. (a) $\frac{1}{a^2}(b^2 - 2ac)$ (b) $-\frac{c}{d}$, (c) $\frac{1}{a^2}(3ad - bc)$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	C	A	D	A	A	A	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	C	D	B	A	D	D	A	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	A	B	D	B	B	B	C	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	A,C,D	C,D	A,D	C,D	B,C	A,B,C	B,D	A,B,C	B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,D	A,B	A,B,D	A,B,D	B,D	B,C,D	A,B,D	A,B,C,D	A,D	A,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	D	B	A,C	B,C,D	A,B,C,D	3.00	2.00	A	
Q.10	A	B	C	D						
	p,r,s	q,s	q,s	p,r,s						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	254	191	24	5	7	4	4	3	3	20

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	1	1	2	3	3	1	2	4
Que.	11	12	13	14	15	16	17	18		
Ans.	8.00	3	4	1	C	D	3	25		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	
Ans.	2	B	C	B	A,D	C	C	D	4	